databricks

Introduction to Data Engineering in Databricks

LECTURE

What is Lakeflow Jobs?

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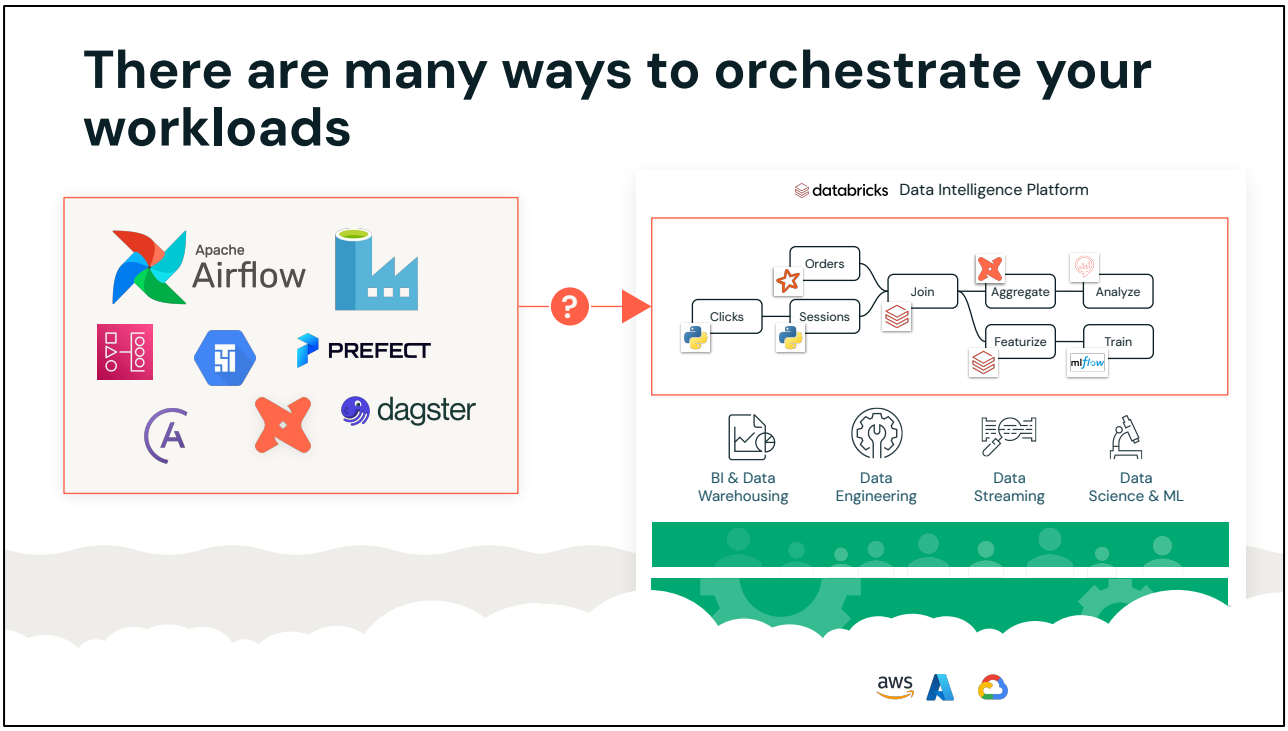
Now let's dive deeper into understanding exactly what Lakeflow Jobs are and why they matter.



This diagram captures a fundamental challenge in modern data architecture: choosing the right orchestration approach for lakehouse workloads. The left panel shows multiple options including open source solutions (Apache Airflow, Prefect, Dagster, dbt), cloud-native services (AWS, Azure, Google Cloud), and custom in-house frameworks.

Diagram on right illustrates a typical data workflow with multiple steps: ingesting sessions and clicks data, joining them, performing featurization and aggregation, analysis, and training models. Then you have various downstream uses including BI & Data Warehousing, Data Streaming, and Data Science & ML.

The question mark here represents a critical challenge: there are many ways to orchestrate these workloads, but which approach is best?

External orchestrators create challenges

Hard to use for many practitioners	Difficult to understand root cause when issues occur	Complex architecture to manage and maintain
Data teams are less productive	Bad data lowers value of downstream applications	Higher cost of ownership and lower reliability

These tools are not unified with your Lakehouse

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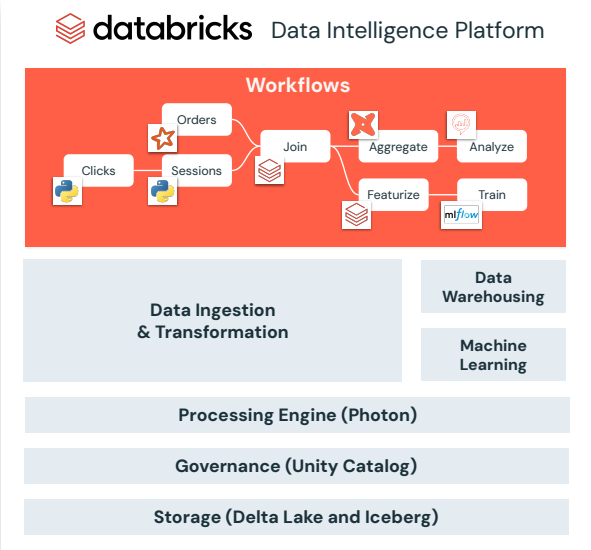
Many organizations use external orchestration tools, but this creates significant challenges. Data teams become less productive because these tools are hard to use for many practitioners. Bad data quality lowers the value of downstream applications.

You also face higher costs of ownership and lower reliability. When issues occur, it's difficult to understand root cause. The complex architecture becomes hard to manage and maintain.

Most importantly, these external tools are not unified with your Lakehouse, creating integration challenges and data silos.

What is Lakeflow Jobs?



Jobs

Unified orchestration for data, analytics, and AI on the Data Intelligence Platform

- Simple authoring
- Actionable insights
- Proven reliability

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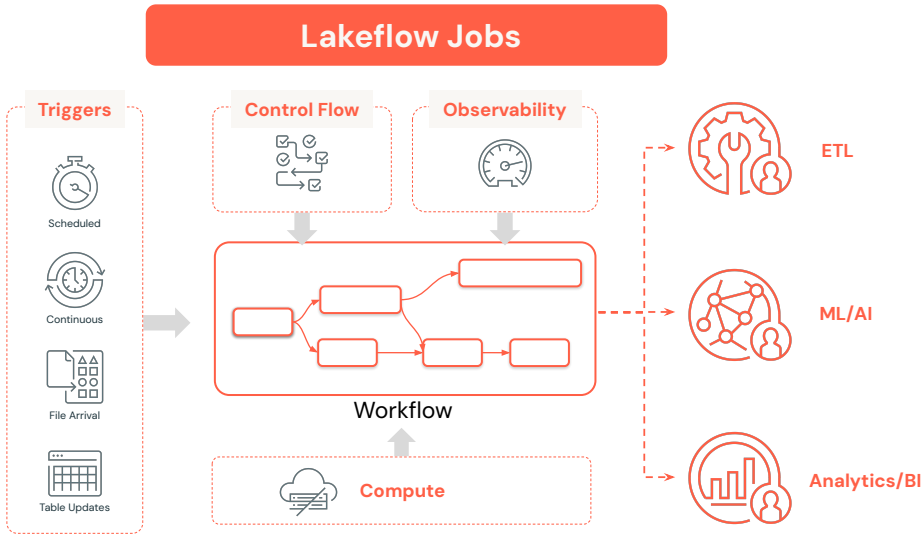
This is where Lakeflow Jobs comes in. It provides unified orchestration for data, analytics, and AI workloads directly on the Data Intelligence Platform.

The key benefits are simple authoring, actionable insights, and proven reliability. Because it's native to the platform, it integrates seamlessly with data ingestion & transformation, the processing engine (Photon), governance (Unity Catalog), storage (Delta Lake), data warehousing, and machine learning capabilities.

The workflow shown here - from sessions and clicks through join, featurize, aggregate, analyze, and train - all runs natively within the same platform, eliminating the integration challenges of external tools.



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- This slide shows the complete architecture of Lakeflow Jobs. At the center is the Workflow engine that coordinates everything.
- The compute layer supports various workload types: ETL, ML/AI, and Analytics/BI operations.
- You have multiple trigger types: Scheduled (time-based), Continuous (always-running), File Arrival (event-driven), and Table Updates.
- Two critical components support the entire system: Observability for monitoring and troubleshooting, and Control Flow for managing task dependencies and execution order.

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Thank you for completing this lesson and continuing your journey to develop your skills with us.